KINGDOM OF CORPUS · STUDY NOTES

Land 12: Pulmora Highlands

The respiratory system · OpenStax Ch. 22

Key ideas

Conducting zone

- Nose, pharynx, larynx, trachea, bronchi and bronchioles down to terminal bronchioles conduct air but do not exchange gas.
- The nasal conchae and mucosa warm, humidify and filter air.
- Pharynx regions: nasopharynx (air only), oropharynx and laryngopharynx (air and food).
- Larynx: thyroid cartilage, cricoid cartilage, epiglottis that covers the airway during swallowing, vocal folds.
- Trachea has C shaped cartilage rings and pseudostratified ciliated columnar epithelium with goblet cells forming the mucociliary escalator.

Respiratory zone and alveoli

- Respiratory bronchioles, alveolar ducts and alveoli exchange gas.
- Type I alveolar cells are thin squamous cells forming most of the wall; type II cells secrete surfactant; alveolar macrophages remove debris.
- Surfactant lowers surface tension so alveoli do not collapse; premature infants may lack it.
- Respiratory membrane: alveolar wall, fused basement membranes, capillary wall, about 0.5 micrometer thick.
- Visceral pleura covers the lung, parietal pleura lines the cavity, and pleural fluid holds them together.

Mechanics of breathing

- Boyle's law: at constant temperature, pressure and volume are inversely related.
- Inspiration: diaphragm and external intercostals contract, thoracic volume rises, intrapulmonary pressure falls below atmospheric.
- Quiet expiration is passive elastic recoil; forced expiration uses internal intercostals and abdominal muscles.
- Intrapleural pressure stays about 4 mm Hg below intrapulmonary pressure; equalizing it causes pneumothorax and collapse.
- Compliance falls in fibrosis and rises in emphysema; airway resistance depends mainly on bronchiole diameter.

Lung volumes and ventilation

- Typical values: TV 500 mL, IRV 3,100 mL, ERV 1,200 mL, RV 1,200 mL.
- VC equals TV plus IRV plus ERV, about 4,800 mL; TLC adds RV, about 6,000 mL.
- IC equals TV plus IRV; FRC equals ERV plus RV.
- Minute ventilation equals TV times rate, about 6,000 mL per minute at rest.
- Alveolar ventilation equals (TV minus dead space of about 150 mL) times rate, about 4,200 mL per minute.

Gas laws and partial pressures

- Dalton's law: each gas exerts pressure in proportion to its percentage; air PO2 is about 160 mm Hg at sea level.
- Henry's law: dissolved gas depends on partial pressure and solubility; CO2 is about 20 times more soluble than O2.
- Alveolar air: PO2 about 104, PCO2 about 40 mm Hg.
- Blood returning to lungs: PO2 about 40, PCO2 about 45 mm Hg; tissues are similar.
- External respiration occurs in the lungs, internal respiration in the tissues.

Ventilation and perfusion matching

- Low alveolar O2 constricts local pulmonary arterioles, diverting blood to well ventilated alveoli.
- High alveolar CO2 dilates local bronchioles to improve ventilation.
- Ventilated alveoli without blood flow act as dead space.
- Blood flowing past unventilated alveoli is a shunt that lowers arterial PO2.

Oxygen transport

- About 98.5 percent of O2 rides on hemoglobin; about 1.5 percent is dissolved.
- The dissociation curve is S shaped and flat above a PO2 of about 60 mm Hg, so saturation stays high.
- Right shift (less affinity, more unloading): higher temperature, CO2, acid (Bohr effect) and BPG.
- Left shift (more affinity): lower temperature, alkalosis, low BPG and fetal hemoglobin.

Carbon dioxide transport

- About 7 to 10 percent dissolved, about 20 percent as carbaminohemoglobin, about 70 percent as bicarbonate ion.
- Carbonic anhydrase in red cells combines CO2 and water into carbonic acid, which splits into hydrogen ion and bicarbonate.
- Chloride shift: chloride enters red cells as bicarbonate leaves in systemic capillaries.
- Haldane effect: deoxygenated hemoglobin carries more CO2 and hydrogen ion.

Control of breathing and disorders

- The medulla's dorsal and ventral respiratory groups set rhythm; the pontine respiratory group smooths transitions.
- Central chemoreceptors respond to CO2 via CSF pH, the strongest everyday drive; peripheral chemoreceptors sense PO2 below about 60 mm Hg.
- Hyperventilation lowers PCO2 and raises pH; hypoventilation does the reverse.
- Asthma: reversible bronchoconstriction and inflammation. COPD: emphysema and chronic bronchitis. Pneumonia: fluid filled alveoli. Cystic fibrosis: faulty CFTR chloride channel and thick mucus.
- Exercise raises ventilation through neural input with near normal blood gases; altitude acclimatization adds hyperventilation, more BPG and erythropoietin driven red cell production.

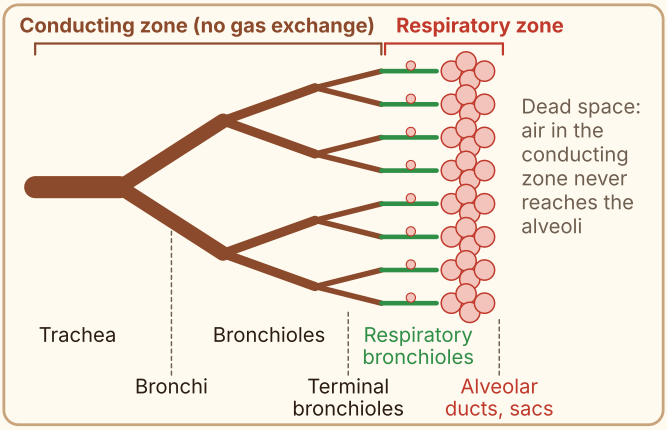
Villager wisdom

- Up here the air is thin, but trust your hemoglobin. Its curve is flat at the top, so it stays nearly full even when oxygen pressure drops a bit.

- Boyle's law rules every breath: make the chest bigger and the pressure inside falls, so air rushes in to fill it.
- Surfactant from type II cells lowers surface tension so the tiny alveoli do not collapse like wet soap bubbles.
- Most carbon dioxide travels as bicarbonate ion, made inside red cells by carbonic anhydrase. Chloride steps in as bicarbonate steps out.
- Your urge to breathe comes mostly from rising carbon dioxide, sensed by chemoreceptors in the medulla, not from falling oxygen.

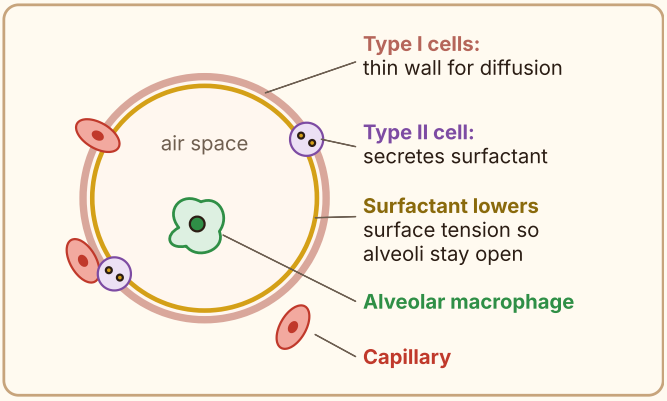
Diagrams

Conducting and respiratory zones



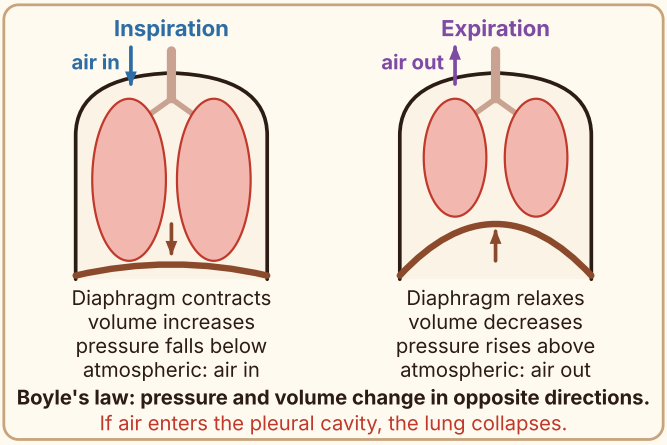
The conducting zone (trachea to terminal bronchioles) only moves air and is anatomic dead space; gas exchange begins at the respiratory bronchioles.

Alveolus and surfactant



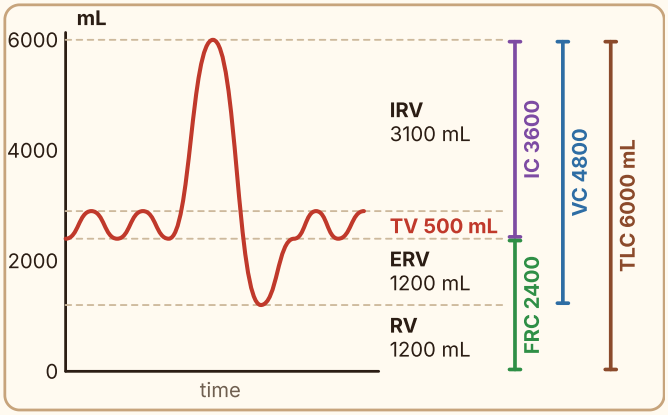
Type I cells form the thin gas exchange wall; type II cells secrete surfactant, which lowers surface tension so alveoli do not collapse.

Boyle's law and breathing



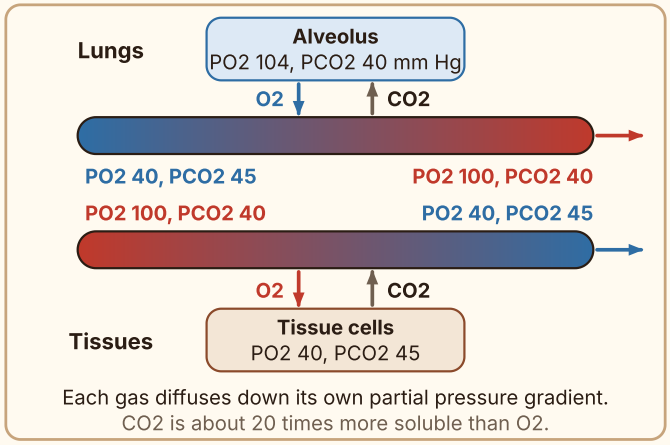
Volume and pressure change in opposite directions: when the diaphragm contracts the chest enlarges, lung pressure drops below atmospheric, and air flows in.

Lung volumes and capacities



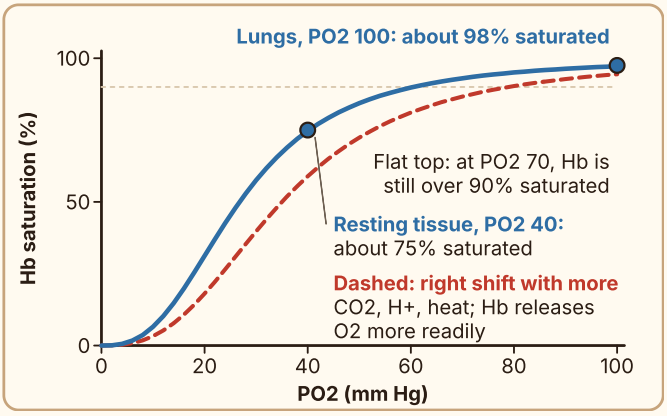
Capacities are sums of volumes: vital capacity = TV + IRV + ERV, and residual volume cannot be measured by spirometry because it never leaves the lungs.

Gas exchange and partial pressures



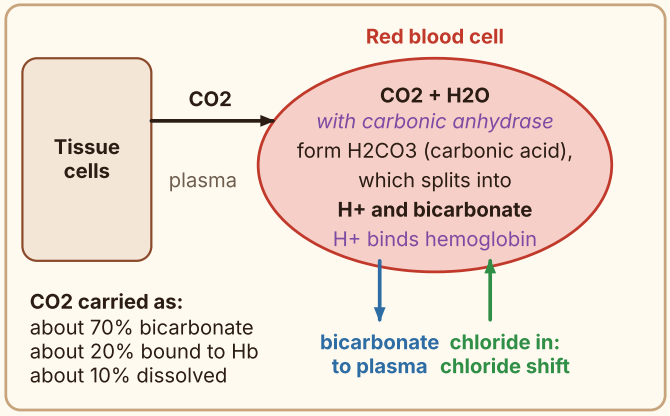
Each gas diffuses from high to low partial pressure: O2 moves from alveoli (104 mm Hg) into blood (40) and from blood into tissues; CO2 moves the opposite way.

Oxygen hemoglobin saturation curve



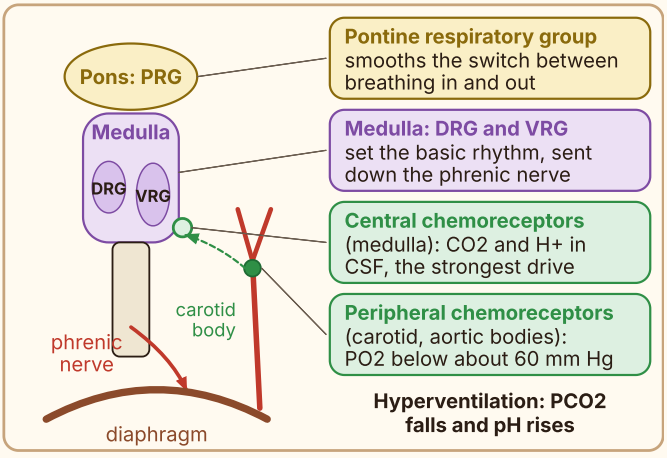
The curve is flat at high PO2, so hemoglobin stays over 90 percent saturated even at 70 mm Hg, and steep at tissue PO2; a right shift means O2 is released more readily.

Carbon dioxide transport



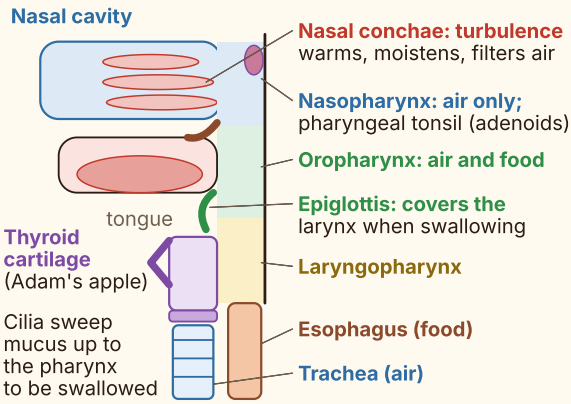
Most CO2 travels as bicarbonate: carbonic anhydrase in red cells makes carbonic acid, which splits into H+ (buffered by hemoglobin) and bicarbonate, which leaves as chloride enters (chloride shift).

Control of breathing



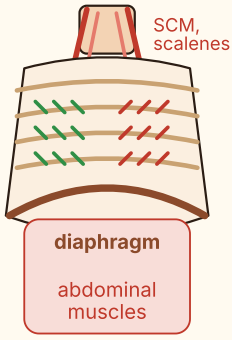
The medulla sets the breathing rhythm and the pons smooths it; central chemoreceptors respond mainly to rising CO2 (as H+ in CSF), and peripheral chemoreceptors respond when PO2 falls below about 60 mm Hg.

The upper airway



Nasal conchae swirl air so it is warmed, moistened and filtered; the nasopharynx carries only air, the oropharynx and laryngopharynx carry air and food, and the thyroid cartilage forms the Adam's apple.

Muscles of breathing



green: external intercostals
red: internal intercostals

Quiet inspiration
diaphragm (most work) plus external intercostals

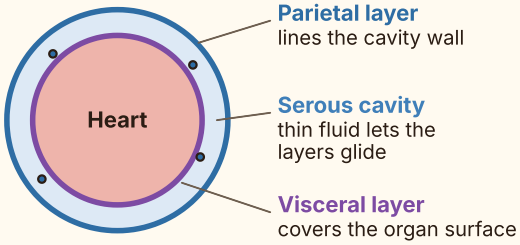
Forced inspiration
adds scalenes, sternocleidomastoid, pec minor

Quiet expiration
passive: elastic recoil of lungs and chest wall

Forced expiration (cough)
internal intercostals and abdominal muscles

The diaphragm does most of the work of quiet inspiration and quiet expiration is passive elastic recoil; forced expiration, as in a cough, uses the internal intercostals and abdominal muscles.

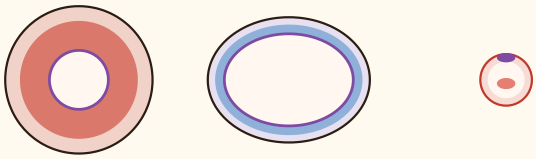
Serous membranes



Pericardium: heart Pleura: lungs Peritoneum: abdomen

A serous membrane has a visceral layer on the organ and a parietal layer on the cavity wall; serous fluid between them lets the layers glide without friction.

Arteries, veins and capillaries



Artery
thick muscular wall, high pressure

Vein
thin wall, large lumen, valves

Capillary
wall one cell thick: site of exchange

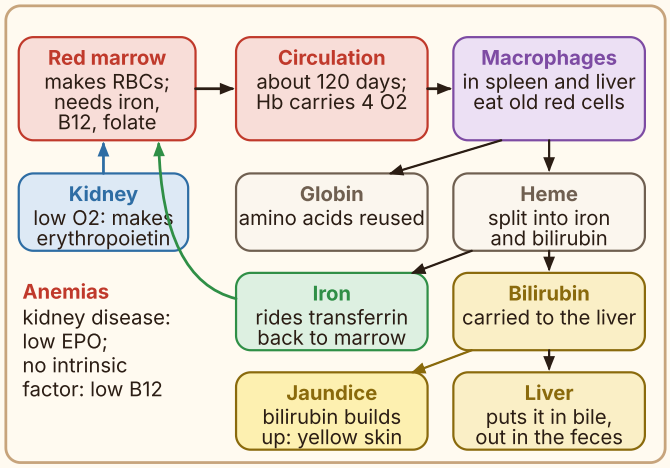


valves stop backflow

Arterioles are the main resistance vessels. Halving the radius raises resistance about 16 times.

Arteries have thick muscular walls, veins have thin walls, large lumens and valves, and capillaries are one cell thick; arterioles set most of the resistance.

Life cycle of a red blood cell



Red cells circulate about 120 days; macrophages recycle them, iron returns to the marrow on transferrin, and bilirubin goes to the liver and out in bile, so a buildup of bilirubin causes jaundice.

Self check

1. **As hemoglobin unloads O_2 in systemic capillaries, it can carry more CO_2. This is called the:**
    A. Haldane effect
    B. Bohr effect
    C. Chloride shift
    D. Henry's law
2. **What is the main function of the thin type I alveolar cells?**
    A. Secrete pulmonary surfactant
    B. Form a thin surface for gas diffusion
    C. Engulf dust and microbes
    D. Produce mucus to trap debris
3. **An anxious patient hyperventilates for several minutes. What happens to arterial PCO_2 and pH?**
    A. PCO_2 rises and pH falls
    B. PCO_2 falls and pH rises
    C. Both PCO_2 and pH rise
    D. Both PCO_2 and pH fall
4. **At altitude, a hiker's arterial PO_2 falls from 100 to 70 mm Hg, yet hemoglobin stays over 90% saturated. Why?**
    A. Hemoglobin binds CO_2 in place of O_2
    B. Plasma dissolves extra oxygen
    C. The curve is flat at high PO_2 values
    D. Each heme binds two O_2 molecules at altitude
5. **What is the function of the C shaped cartilage rings in the trachea?**
    A. Keep the airway open
    B. Close the airway during swallowing
    C. Secrete mucus onto the lumen
    D. Allow gas exchange with the blood
6. **Which receptors respond strongly when arterial PO_2 falls below about 60 mm Hg?**
    A. Central chemoreceptors in the medulla
    B. Peripheral chemoreceptors in carotid and aortic bodies
    C. Stretch receptors in the bronchioles
    D. Baroreceptors in the carotid sinus
7. **As bicarbonate ion leaves red blood cells in systemic capillaries, which ion moves in to balance the charge?**
    A. Sodium ion
    B. Hydrogen ion
    C. Potassium ion
    D. Chloride ion

8. **A patient hypoventilates and arterial PCO2 rises. Which receptors respond most strongly to restore breathing?**
    A. Irritant receptors in the airway
    B. Central chemoreceptors in the medulla
    C. Baroreceptors in the aortic arch
    D. Stretch receptors in the bronchioles
9. **Which muscle does most of the work of quiet inspiration?**
    A. Internal intercostals
    B. Rectus abdominis
    C. Sternocleidomastoid
    D. Diaphragm
10. **In actively exercising muscle, the oxygen hemoglobin dissociation curve shifts right. What does this mean?**
    A. Hemoglobin holds O2 more tightly
    B. Hemoglobin releases O2 more readily
    C. Blood carries less CO2
    D. Saturation in the lungs drops sharply
11. **Given TV 500 mL, IRV 3000 mL, ERV 1100 mL and RV 1200 mL, what is the vital capacity?**
    A. 3,500 mL
    B. 4,600 mL
    C. 5,800 mL
    D. 2,300 mL
12. **Henry's law states that the amount of gas dissolving in a liquid depends on:**
    A. Its partial pressure and its solubility
    B. The total pressure of all gases only
    C. The volume of its container only
    D. The temperature of the air only

Answer key

1. **A.** Haldane effect. Deoxygenated hemoglobin binds CO2 and hydrogen ions more readily, the Haldane effect. The Bohr effect is the reverse view: CO2 and acid lowering O2 affinity.
2. **B.** Form a thin surface for gas diffusion. Type I cells cover most of the alveolar surface and form part of the respiratory membrane. Surfactant is secreted by type II cells, a common mix up.
3. **B.** PCO2 falls and pH rises. Blowing off CO2 shifts the carbonic acid equilibrium away from hydrogen ions, causing respiratory alkalosis. Rising PCO2 with falling pH describes hypoventilation.
4. **C.** The curve is flat at high PO2 values. The top of the S shaped curve is a plateau, so fairly large drops in PO2 there barely change saturation, giving a safety margin. Each heme still binds only one O2 molecule, and dissolved oxygen in plasma stays tiny, so neither explains the high saturation.
5. **A.** Keep the airway open. The cartilage rings prevent collapse, and their open back lets the esophagus expand during swallowing. Closing the airway during swallowing is the role of the epiglottis.

6. **B.** Peripheral chemoreceptors in carotid and aortic bodies. Peripheral chemoreceptors are the only chemoreceptors that sense low O2 directly. Central chemoreceptors respond to CO2 through pH changes in cerebrospinal fluid, not to O2.
7. **D.** Chloride ion. In the chloride shift, chloride ion enters the red cell in exchange for bicarbonate ion so the cell stays electrically neutral. Hydrogen ions are produced inside the cell and buffered by hemoglobin; they are not the exchange partner.
8. **B.** Central chemoreceptors in the medulla. CO2 crosses into cerebrospinal fluid and lowers its pH, which strongly stimulates central chemoreceptors in the medulla to increase the rate and depth of breathing. Stretch receptors guard against overinflation; they do not sense blood gases.
9. **D.** Diaphragm. Contraction of the dome shaped diaphragm enlarges the thoracic cavity vertically. The sternocleidomastoid is an accessory muscle used only in forced inspiration.
10. **B.** Hemoglobin releases O2 more readily. Higher temperature, more CO2, lower pH and more BPG in working muscle lower hemoglobin's affinity for oxygen, so more oxygen unloads where it is needed; the part of this shift caused by CO2 and low pH is called the Bohr effect. A left shift is the one that makes hemoglobin hold oxygen more tightly.
11. **B.** 4,600 mL. Vital capacity is TV plus IRV plus ERV, which is 500 plus 3000 plus 1100, or 4,600 mL. Adding residual volume gives 5,800 mL, which is total lung capacity, not vital capacity.
12. **A.** Its partial pressure and its solubility. Henry's law explains why gas dissolves in plasma in proportion to its partial pressure and solubility. Dalton's law, not Henry's, concerns the total pressure of a gas mixture.

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